Content Planning for Neural Story Generation with Aristotelian Rescoring

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arXiv:2009.09870v2 [cs.CL] 9 Oct 2020

Abstract

Long-form narrative text generated from large language models manages a fluent impersonation of human writing, but only at the local sentence level, and lacks structure or global cohesion. We posit that many of the problems of story generation can be addressed via high-quality content planning, and present a system that focuses on how to learn good *plot structures* to guide story generation. We utilize a plot-generation language model along with an ensemble of rescoring models that each implement an aspect of good story-writing as detailed in Aristotle's *Poetics*. We find that stories written with our more principled plot-structure are both more relevant to a given prompt and higher quality than baselines that do not content plan, or that plan in an unprincipled way.[1]

1 Introduction

Despite many recent advances in Natural Language Generation, successful creative narrative composition remains elusive. Current neural approaches are plagued by difficulty in mastering structure, will veer between topics, and lack long-range cohesion. They successfully imitate the fluency and style of human writing, but on closer inspection sentences do not fit together to form a whole, and the reader is left with the impression that the generation has no content (See et al., 2019). This lack of structure also degrades the relevance of generations conditioned on a prompt or other source text - a strong language model will repeat key phrases from a given prompt but will not remain on topic. These issues are illustrated in the `Naive Generated Story` in Table 1, where many of the sentences individually are fine, but do not fit together as one story, and do not all relate to the prompt.

We hypothesise that this problem can be addressed with a focus on deeper latent narrative structures. In Aristotle's *Poetics*, one of the most enduring treatises on the craft of writing good stories, the philosopher lays out the elements of story in order of importance. They are:

1. event choice and arrangement
2. character
3. relevant content[2]
4. diction

An amateur masters skills later in the list, but mastery of event choice and event arrangement is what distinguishes a good writer (Aristotle). Next is character, then relevance, and only finally do style and diction matter.

This philosophical framework fits remarkably well into the traditional Natural Language Generation Pipeline approach that emphasizes Content Planning (Reiter and Dale, 1997). The pipeline divides generation into three steps: *Content Planning*, *Microplanning* and *Surface Realization*, where at each step input is modified and refined, getting closer to the final textual output. Incorporating a plot in order to generate stories can then be viewed as a proxy for Content Planning/MicroPlanning before a language model makes use of it to convert it to a readable and grammatically correct natural language output (Surface Realization).

Inspired by both the Aristotelian and Content Planning Frameworks, we develop a novel system for story generation. We focus on developing a system that can learn to expertly select events, characters, and relevant content, and write good *plot structures*. After the work on the plot is complete, a large language model can then do what it does best and fill in the descriptions, details, and local specifics of each story.

[1] Code at https://github.com/PlusLabNLP/story-gen-BART

[2] This directly translates as *thought* in the original, but is explained to clearly mean the selection of relevant information

Prompt: Write the most beautiful end to the universe that you can imagine
Naive Plot with Semantic Role Labelling Tags:
<A1> The universe <V> end </s> </s> <A0> ent 0 <V> see <A1> ent 3 </s> </s> <V> dying <A1> ent 1 # <A0> ent 1 <V> left <A1> ent 0 </s> <A0> the last human <V> live </s> <A1> ent 6 <V> end # <A1> ent 2 <V> come </s> <A1> the last one <V> die </s> <A1> a universe of life <V> left
Revised Plot with Semantic Role Labelling Tags:
<A2> The light <V> filled <A2> the sky </s> </s> </s> <A2> A bright flash <V> lit </s> </s> # <V> began <A2> to grow # <A2> the stars <V> grow </s> </s> <V> began <A2> ent 0 <A2> to fade # <A2> The stars <V> fade </s> </s> <A0> ent 2 <V> looked <A2> ent 1 me </s> <V> dying <A2> star
Naive Generated Story:
"You're dead. I'll be dead, and there'll always be no more of you."* *"No, my dear. It's not true!" - The last of the world is dead, * * "It'll never be over. *The universe* will n't end. Not yet. But I'll see it, one way or another. There is no other way."* *My last thought, the *dying* man *left* me. He told that I was *the last human* to live. And he told me the universe would end, and the end will come soon. That your death would be *the last one* to die, and the first one to be the most beautiful. * "There is *a universe of life left*, my dear, and that's all there is.
Generated Story (from revision):
It was the end of the universe. *The light filled the sky*. The galaxy. All the light in all the worlds, and all the stars. *A bright flash*. The first light. And then a bright flash of light... then a flash of darkness. Then a bright light. A flash of blackness and then the stars *began to grow*. It was a bright, beautiful bright, light. And then, slowly the *stars* started to *grow* in number. There were millions and millions of stars.. But, slowly but surely, all the galaxies *began to fade*. Slowly, but surely. Finally, it *looked* at me with its eyes. One *dying star*.
Gold Story:
It slept, and dreamt of twirling lights. It cradled countless twirling lights, all dancing in their flashing ways, dancing to the sound of a faraway flute. The light was vibrant and warm and spun all around It. The tiny lights grew with the tune, and more lights came to dance within the luminescence. It was surrounded by light, all waltzing in their ways to the distant music. It, too, swirled with the swelling symphony, watching the lights. There were lights of all different sizes and shades and brightness, but each was its own, unique light. Each one was brilliant. Each one was beautiful. All dancers to the Song...

Table 1: Our proposed plot and story generation structure. We generate a Naive Plot, revise it with Aristotelian rescorers, then generate a story. In plots, <V> denotes verbs while <A$\{0, 1, 2\}$ > denote arguments. ent $\{0...n\}$ are entities. We removed newline symbol <P> from the generated stories and detokenized for better display.

For plot generation, we employ a few event-choice and event-arrangement rescoring models which assist in building the arc and cohesion of the plot, a character rescoring model that helps select which characters appear where, and a relevance model that is responsible for keeping the plot structure and the story on topic. As both improving plot-generation via rescoring and using an Aristotelian framework for neural generation are novel concepts, there is no previous work on how to implement them in practice.

Our contributions are: **1)** we propose to leverage the principled Aristotelian framework for content planning, **2)** we propose an implementation of the framework using a revision-based approach via several rescoring models **3)** we show strong experimental results against 4 baselines.

2 Background

Existing work in neural story generation has established the strength of adding a content planning stage to structure the generated content (Yao et al., 2019; Fan et al., 2019) (discussed in more detail in Section 7). Specifically, this line of work trains a pipeline with one model that generates from *prompt* $\rightarrow$ *plot* and another that generates from *prompt* + *plot* $\rightarrow$ story. It modifies the standard conditional generation task with a source $x = x_1...x_n$ (in this case, a prompt) and target $y = y_1...y_n$ (in this case, a story) to condition also on an abstract intermediate representation $\mathbf{z}$. Note that the approach is not truly modelling $p(\mathbf{y}|\mathbf{x})$ since that would involve summing over all $\mathbf{z}$. Instead, it models $p(\mathbf{y},\mathbf{z}|\mathbf{x}) = p(\mathbf{z}|\mathbf{x})p(\mathbf{y}|\mathbf{z},\mathbf{x})$, but only shows the generated story $\mathbf{y}$ at inference time.

This is a more controllable task than open-domain generation conditioned on only a prompt $\mathbf{x}$, provided that a good interim structure $\mathbf{z}$ can be learnt. We follow this line and explore ways to improve plot planning to close the gap between stories generated from gold plots and those from model-generated plots.

Plot Representation As there are no large datasets with parallel gold-standard plots and stories, all work on plot generation depends on silver-standard plots extracted from stories. We follow Fan et al. (2019) to represent plots in Semantic Role Labelling (SRL) format. We run coreference resolution to identify entities, and use a compression algorithm to discard less salient information.[3]

[3]Our representation is conceptually quite similar to that of Fan et al. (2019), but is not identical, as their code is un-

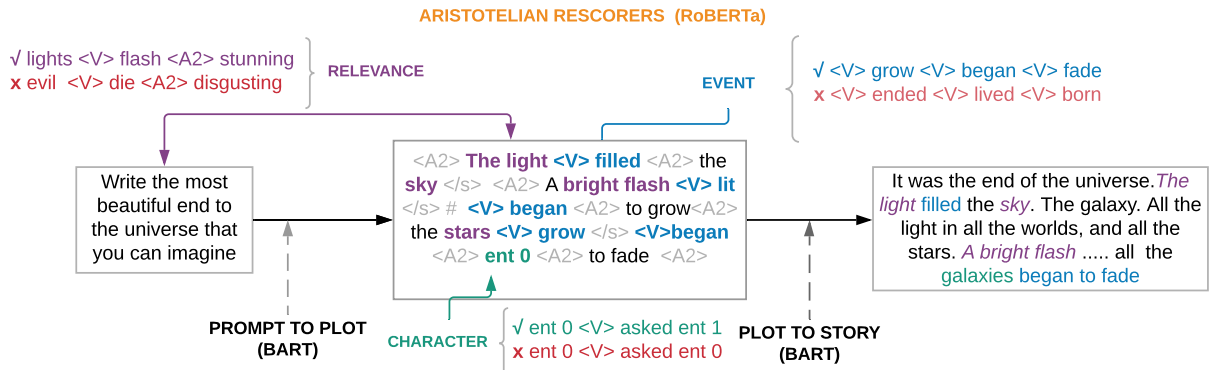

Figure 1: The full story generation pipeline. Base generation models in gray. Aristotelian rescorer input in color, with example positive & negative training pairs. Plot structure shown with SRL tags for verbs and their arguments.

3 Approach

We focus on learning the best interim plot structure between the input prompt and the story surface realisation. As such, we learn the plot model $p(\mathbf{z}|\mathbf{x})$ and the story model $p(\mathbf{y}|\mathbf{z},\mathbf{x})$ by fine-tuning a pre-trained conditional language model BART (Lewis et al., 2019) on 1) pairs of prompts and extracted silver standard plot-structures, and 2) pairs of prompts + plots and stories, respectively. Full implementation details can be found in Appendix A.3.

We propose to modify the decoding objective to incorporate input from each Aristotelian rescoring model $a \in A$ (the complete set of rescoring models, detailed further in Section 3.1) and re-rank the original, or "naive" plot model hypotheses, bringing the plot representation closer to each rescoring model's specialty and desireable story attribute. A diagram of our final system in Figure 1 shows each step of the generation process. The modified decoding objective becomes:

$$f_\lambda(\mathbf{x},\mathbf{z}) = \sum_{i}^{m} -\log p(z|z<i,\mathbf{x}) + \sum_{j}^{|A|} \lambda_j a_j(\mathbf{x}, z_{i\ldots m}) \tag{1}$$

where λ_j is the learned weight of the score given by a_j, as detailed in Section 3.3.

3.1 Aristotelian Rescoring Models

For all of our rescoring models, we train classifiers to *distinguish* positive examples - the silver extracted plots - and negative examples, which are plots that are worse with regard to the aspect that we desire to encode, given the prompt $\mathbf{x}$. The intuition is that if the rescoring model can learn a particular principle, it can assist the plot-generation language model in creating content that encapsulates that principle. Mathematically, the classifiers learn $p(l|\mathbf{x},\mathbf{z}) = \frac{p(\mathbf{x},\mathbf{z},l)}{p(\mathbf{x},\mathbf{z})}$, and we use the probability of the plot being a positive example (a more plausible plot) as our Aristotelian rescoring model:

$$a_j(\mathbf{x}, z_{i\ldots m}) = p_j(l_{pos}|\mathbf{x}, z_{i\ldots m}). \tag{2}$$

What differs for each model a_j that specialises in a different Aristotelian aspect is the set of negative examples that we generated to capture the type of information it has learnt to discriminate, and the features it learns. We give more details about each Aristotelian rescorer as follows. Example simplified positive and negative pairs for each are depicted in Figure 1. Full unsimplified pairs of positive and negative examples for each rescorer can be found in Appendix B.

Event Rescorers. The SRL extracted plots provide us with a structure that is very similar to event representations in event extraction literature. We thus consider an *Event* to be composed of an action-based verb and its subject and object (a verb, subject, object tuple).[4] We experiment with three different ways to construct positive and negative event examples. SRL based plots are structured, and a random shuffle would be trivial to distinguish, so we need more nuanced ways to learn good event choice and arrangement. We try:

- **inter-sentence shuffled events** we permute all sentences as a full chunk, and keep all events within a sentence together.
- **intra-sentence shuffled events** we permute the event tuples within a sentence, but keep each verb and its arguments together.
- **verb-shuffled events** we permute only the event verbs within a sentence, leaving their arguments and contexts unchanged.

available. We used AllenNLP (Gardner et al., 2018) to run the SRL model (He et al., 2017) and Co-reference model (Lee et al., 2017) and determined our own compression algorithm experimentally. Further details in Appendix A.5

[4] With modal verbs generally excluded in a stop list, which is included in Appendix A.5